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Paleoposition of the northern margin of Armorica in Late Devonian times: Paleomagnetic and rock magnetic results from the Frankenstein Intrusive Complex (Mid-German Crystalline Rise)

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Abstract. A detailed rockmagnetic and paleomagnetic study of the Frankenstein Intrusive Complex (FIC, 363 ± 7 Ma in age), southwestern Germany, has been carried out in order to test whether the paleomagnetic declinations within the internal parts of the European Variscides follow a common trend controlled by the structural grain of the orogen or whether the distribution of declinations follows a random pattern. In addition, new paleomagnetic data will contribute to further refine the drift history of the Armorican Terrane Assemblage during late Paleozoic times. The unmetamorphosed Frankenstein Igneous Complex forms part of the Mid-German Crystalline Rise within the Armorican Terrane Assemblage, amalgamated to Avalonia and Baltica during the Variscan orogeny. Stepwise thermal and alternating field (AF) demagnetization experiments identify four components (A, B, C_n, and C_r) of magnetization. Component A is a recent viscous overprint that parallels the present-day geomagnetic field. The high unblocking temperatures of A in some samples can be explained by the presence of multidomain magnetite. Component B is carried by hematite and is considered to be a late Carboniferous remagnetization. Components C_r and C_n, found in 110 samples from 20 sites, are antiparallel and pass a class C reversal test. Their primary origin is further supported by a contact test. C_r and C_n have maximum unblocking temperatures of up to 580°C and occasionally above over 600°C, indicating magnetite and hematite to carry the characteristic remanent magnetization. Ore microscopy revealed that primary hematite lamellae in ilmenite carry C_r and C_n and that a secondary generation of hematite is the carrier of B. The overall site mean direction of C_r and C_n (20 sites) based on endpoints and remagnetization circle analysis of 198°/40° (declination/inclination) with a 95% confidence limit (α_{95}) of 4.9° and a precision parameter (k) of 45.0 defines a paleopole at 15°S, 9°W. When comparing to the apparent polar wander path for stable Europe this result indicates up to 27° counterclockwise rotation of the FIC region. This is significantly less than reported previously for the Odenwald, Spessart, Black Forest, and the Vosges Massifs of the central European Variscides. Furthermore, the resulting paleolatitude indicates significant separation between Baltica and this sector of the Armorican Terrane Assemblage in the latest Devonian.

1. Introduction

Previous paleomagnetic results and geological evidence indicate that the Rheic ocean, separating Laurentia and Baltica to the north and the Armorican Terrane Assemblage to the south, was already closed in Late Devonian time [*Tait et al.*, 1997, and references therein]. However the paleodeclinations obtained from Late Devonian and Carboniferous rocks in the Variscan Fold Belt show large variations. Close to the northern thrust front of the orogen, declination changes seem to be controlled by changes in regional strike and have been interpreted to reflect oroclinal bending [*Bachtadse and Van der Voo*, 1986]. A recent paleomagnetic study by *Tait et al.* [1996] on sedimentary rocks from the Moravo-Silesian zone further supports this interpretation. Their results confirm earlier interpretations that the Moravo-Silesian zone originally formed an integral part of the Rhenohercynian zone of the European Variscides that has subsequently been bent by 90° clockwise around the Bohemian Massif. Restoring the Moravo-Silesian zone to its original Devonian orientation brings both its structural trend and paleodeclinations into agreement with the Rhenohercynian where the paleodeclinations show only minor rotations [*Bachtadse et al.*, 1983]. Whereas the bending of the Variscides around the Bay of Biscay has been attributed to the northward translation of the Ebro-Pyrenean microplate into the foldbelt [*Bachtadse and Van der Voo*, 1986], the northeastward translation and indentation of the Bohemian Massif into the Rhenohercynian has been postulated to be responsible for oroclinal bending in the eastern sector of the foldbelt [*Tait et al.*, 1996].

However, a series of paleomagnetic studies on Devonian and Carboniferous metamorphic, intrusive and extrusive rocks in the more internal parts of the orogen (Massif Central, Vosges, Black Forest, Pfalz, Odenwald and Spessart) yield paleodeclinations that are not in agreement with simple oroclinal bending as outlined above but have been interpreted to be a result of succes-

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sive, large-scale clockwise and counterclockwise rotations of individual blocks within the orogen [*Edel*, 1987; *Edel and Schneider*, 1995; *Edel and Wickert*, 1991].

To test the above hypothesis and to further refine the drift history of the Armorican Terrane Assemblage, a detailed paleomagnetic and rock magnetic study was undertaken on the Frankenstein Intrusive Complex (FIC) of Late Devonian age in the crystalline Odenwald of southwest Germany.

2. Geological Setting

The FIC and its country rocks form the northernmost structural unit of the crystalline Odenwald and form part of the Mid-German Crystalline Rise (Figure 1). As previously defined by *Kossmat* [1927], the Mid-German Crystalline Rise separates the Rhenohercynian and Saxothuringian Zones of the mid-European Variscides. In more recent plate tectonic concepts of the European Variscides, the Mid-German Crystalline Rise is thought to represent the relicts of several magmatic arcs on the southern edge of the Rhenohercynian (Avalonia) and the northern edge of the Saxothuringian tectonostratigraphic units (Armorica), which were active during the closure of the Ordovician-Silurian Rheic and the Devonian-Carboniferous Rhenohercynian oceans [*Oncken*, 1997 and references therein].

The crystalline Odenwald can be divided into four units (Figure 2a) with differing tectonostratigraphic evolution [*Krohe*, 1992]. During the Devonian and Carboniferous, plutons of various compositions intruded into lower Paleozoic metasedimentary and magmatic rocks of units 1-3 in Figure 2a. These magmatic rocks are characterized by diorites and gabbroic cumulates in unit 1, basic to intermediate granitoids in unit 2, and intermediate to acidic granitoids in unit 3. In this region the crustal thickness increases from 20 in the north to 60 km in the south as intrusion depths of the plutons increase from 8 to 15 km [*Henes-Klaiber*, 1992]. The diorites and gabbros of unit 1 intruded in the Late Devonian [*Kirsch et al.*, 1988], whereas granitoids of units 2 and 3 were emplaced during the early Carboniferous (340-324 Ma [*Kreuzer and Harre*, 1975]). Units 1-3 formed a magmatic arc on the northern edge of Armorica and were linked to southward subduction and underthrusting of the Rhenohercynian under the Saxothuringian zone [*Oncken*, 1997].

In unit 4 the upper amphibolite facies orthogneisses [*Willner et al.*, 1991] have a different tectonostratigraphic history. During the Late Silurian to Early Devonian, their educts intruded into micaschists and paragneisses [*Lippolt*, 1986] and formed part of an active magmatic arc during the closure of the Rheic ocean. The large-scale folding of thrust stacks in this unit is interpreted to be the result of accretion by underplating of continental crust of the lower Rhenohercynian plate to the base of the upper Saxothuringian plate during the final oblique collision of the Armorican Terrane Assemblage and Avalonia in the lower Carboniferous [*Oncken*, 1997].

Within this geodynamic framework the FIC (Figure 2b), part of unit 1 (Figure 2a), represents the earliest unmetamorphosed magmatic member in the crystalline Odenwald. The concordance between $^{40}Ar/^{39}Ar$ isotope ages on hornblende of 363 ± 7 Ma and plagioclase of 359 ± 3 Ma and $^{207}Pb/^{206}Pb$ isotope ages on zircons of 362 ± 9 Ma demonstrates that the FIC cooled relatively rapid in <10 Myr to below the 200°C closure temperature of plagioclase in the K-Ar system [*Kirsch et al.*, 1988]. Furthermore, no resetting of the isotope systems has been observed, thus ruling out any post-Devonian reheating of the FIC above 200°C. Except for its western rim where the FIC is cut off by the eastern flank of the Rhine Graben, the FIC is surrounded by medium-grade metamorphic rocks of lower Paleozoic age. A sinistral shear zone with orthogneisses marks the boundary be-

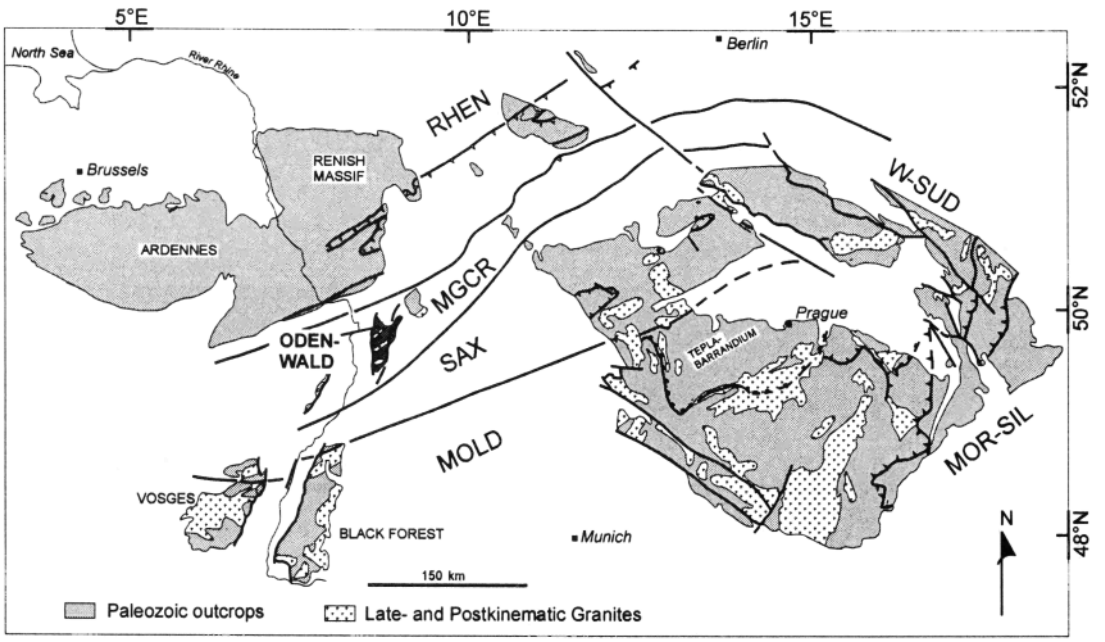


Figure 1. The mid- and east European Variscides with their tectonostratigraphic units and the location of the crystalline Odenwald. RHEN, Rhenohercynian (southern margin of Avalonia); MGCR, Mid-German Crystalline Rise (northern margin of Armorica); SAX, Saxothuringian; MOLD, Moldanubian; W-SUD, West-Sudetes; MOR-SIL, Moravo-Silesian. After *Dallmeyer et al.* [1995], copyright by Springer-Verlag, New York, with permission from the publisher and author.

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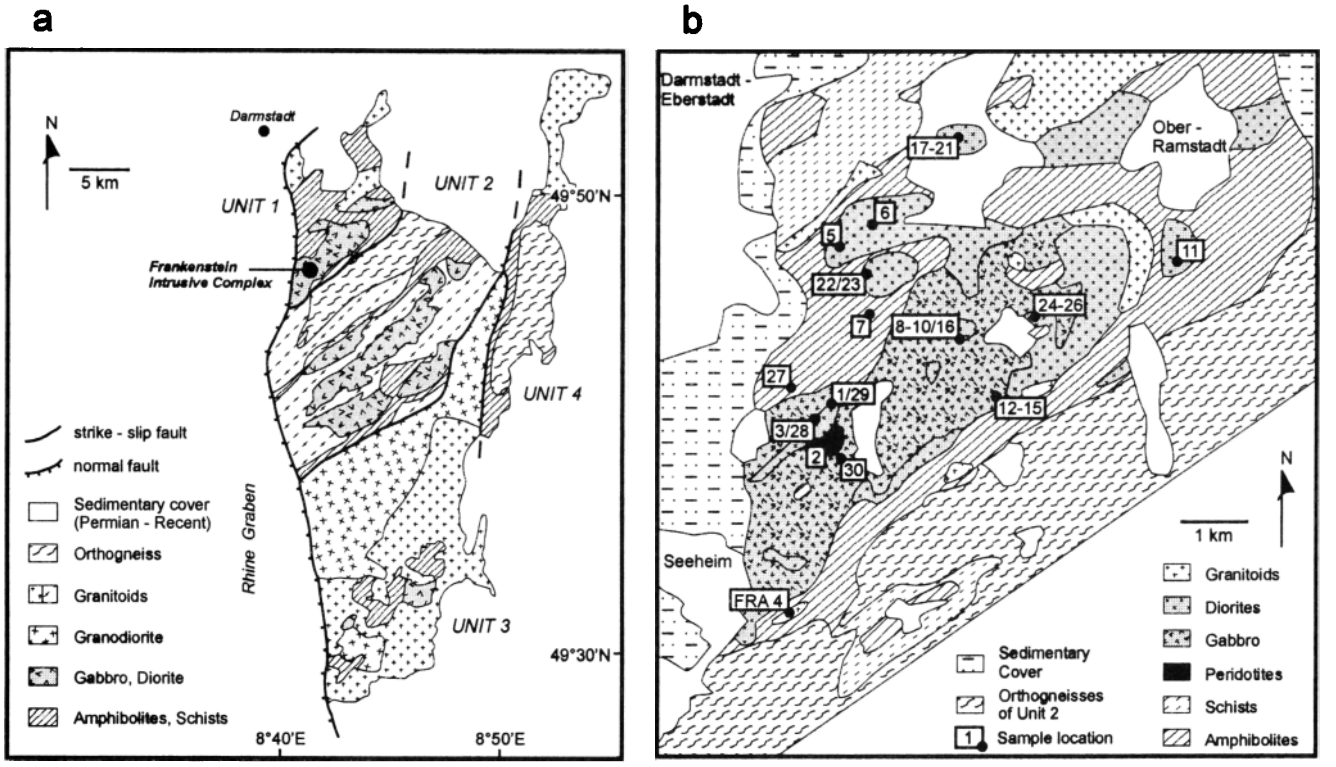


Figure 2. (a) Geological sketch map of the crystalline Odenwald. Paleomagnetic sampling was restricted on the Frankenstein Intrusive Complex (FIC) in unit 1. (b) Geological map of the FIC with location of the paleomagnetic sampling sites. Information for both figures is taken from *Nickel* [1985].

tween units 1 and 2 of the crystalline Odenwald [*Krohe*, 1992]. The composition of the FIC ranges from peridotites through gabbros and diorites to granitoids. Several NW-SE trending aplitic dykes cut the pluton. In two quarries (sites 8-10 and 12-15) magmatic cumulates of light-colored, plagioclase-rich and dark-colored, pyroxene-rich layers were identified. Gravitational crystal sagging or diffusion-nucleation processes are thought to be responsible for this magmatic layering [*Kreher*, 1994], which forms subhorizontal, undulating surfaces. The distance to the margin of the intrusion is sufficient (>1km) to rule out steep or vertical cooling-related crystallization surfaces as the cause of this layering. Together with homogenous metamorphic conditions in the country rocks surrounding the pluton [*Willner et al.*, 1991], the subhorizontal layering provides strong evidence that the FIC was not significantly tilted about a horizontal axis after emplacement. The mean orientation of the magmatic layering is in good agreement with the bedding of younger Permian sediments directly north of the FIC, which dip 5°E. Therefore, only minor Tertiary tilting during formation of the Rhine Graben is considered relevant for this study, and the above value has been used for tilt correction.

3. Sampling

Paleomagnetic sampling was concentrated on the FIC, where well-defined cooling ages and good structural control from magmatic layering are available. Other gabbroic and dioritic suites of units 2 and 3 of the crystalline Odenwald were not sampled owing to the lack of clearly identifiable structural attitudes and insufficiently precise age control.

Cores of 2.54 cm diameter were drilled from unweathered exposures of dioritic and gabbroic rocks of the FIC in quarries and road cuts. Amphibolites were sampled at sites 7 and 27 along the northwest boundary of the FIC (Figure 2b) but did not yield interpretable paleomagnetic results. An aplitic dyke, thought to be Permo-Carboniferous in age and cutting the gabbro at site 16 (Figure 2b), was sampled for a contact test. In total, six to nine cores were collected at each of the 30 sites and cut into standard 2.2 cm long specimens for paleomagnetic measurements.

4. Paleomagnetic Methods and Results

The specimens were subjected to stepwise thermal demagnetization with two Schonsted TSD-1 furnaces or alternating field (AF) demagnetization using a 2G Enterprises AF demagnetizer. Measurements of remanent magnetization were made with a 2G Enterprises DC-SQUID magnetometer. All equipment used for the paleomagnetic measurements is housed in a magnetically shielded room. Although thermal and AF treatments yield similar results, thermal demagnetization was judged superior and most specimens subjected to stepwise thermal demagnetization up to 700°C. To minimize the effects of thermal gradients in the furnaces, each specimen was heated in the same furnace in the same position. The stepwise demagnetization results were analyzed using orthogonal vector plots [*Zijderveld*, 1967] and stereographic projections. Linear segments and remagnetization circles trajectories were identified by eye and subjected to principal component analysis [*Kirschvink*, 1980]. Analysis of stable endpoints and remagnetisation circles was

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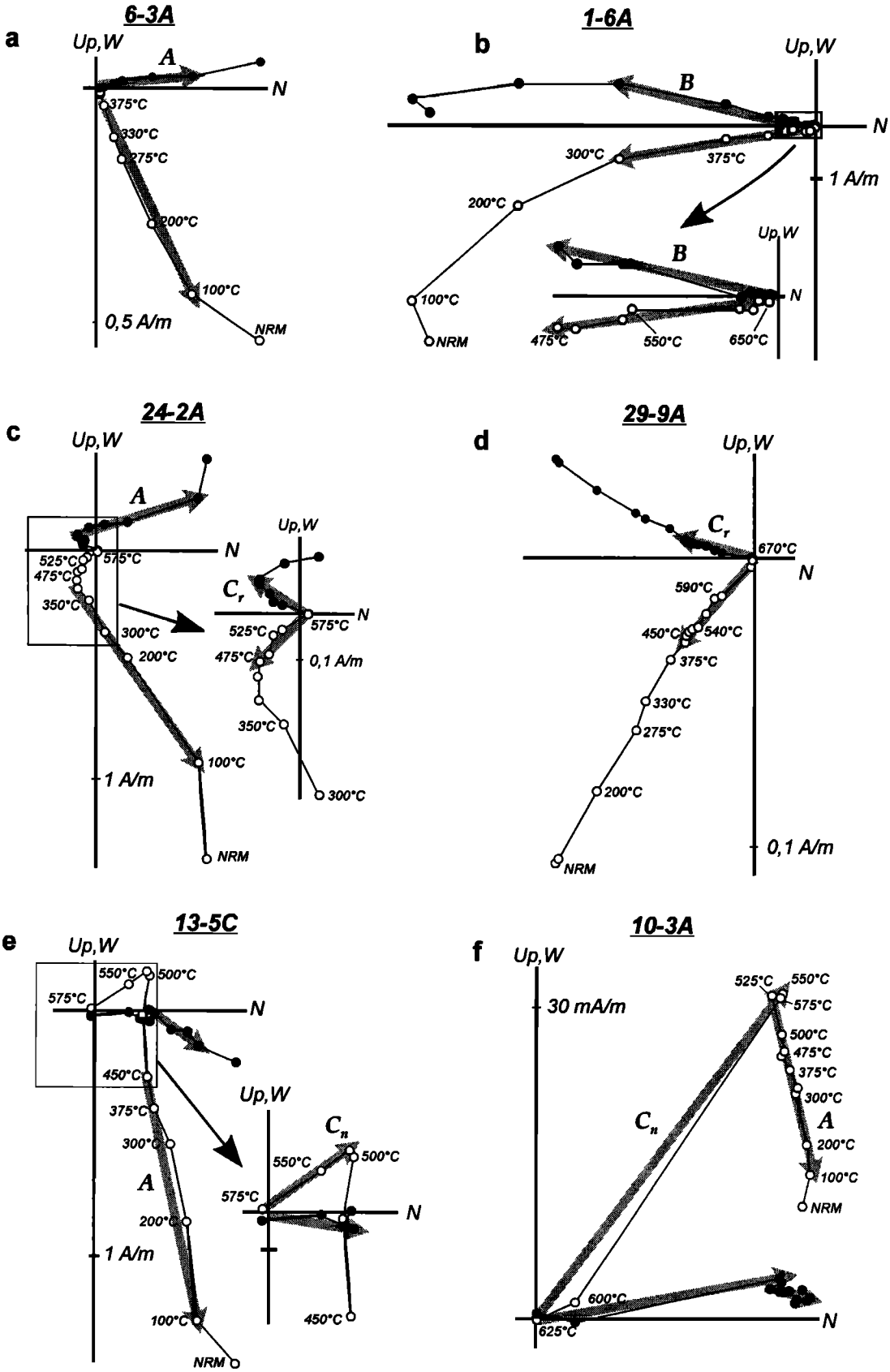


Figure 3. (a-f) Thermal demagnetization behaviour of six typical specimens in orthogonal projection (open and solid circles are vertical and horizontal planes, respectively) showing remanence components A, B, C_r, and C_n.

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performed using the algorithm given by *McFadden and McElhinny* [1988]. Mean site directions were calculated using *Fisher* [1953] statistics. All demagnetization diagrams are shown in geographic coordinates.

Four distinct components of magnetization (A, B, C_r, and C_n) are identified in the sample set. After removal of viscous remanence by heating up to 100°C, A was identified in 75% of the specimens where it usually reached maximum unblocking temperatures (T_{UB}) of 400°C. In few samples, A was the only component identified (Figure 3a), and it was occasionally stable up to 450°C or even higher temperatures (Figure 3f). The resulting unit mean direction for component A is close to the direction of the Earth's present-day field in the sampling area, and so it is interpreted to be of recent origin.

Three additional components, B, C_r, and C_n, were observed at higher temperatures between 475° and 670°C. B was identified in 20 specimens from four sites and was usually stable above 650°C, indicating hematite is its carrier (Figure 3b). Its southerly declination and shallow positive inclination is similar to magnetization directions previously identified in Carbonifer-

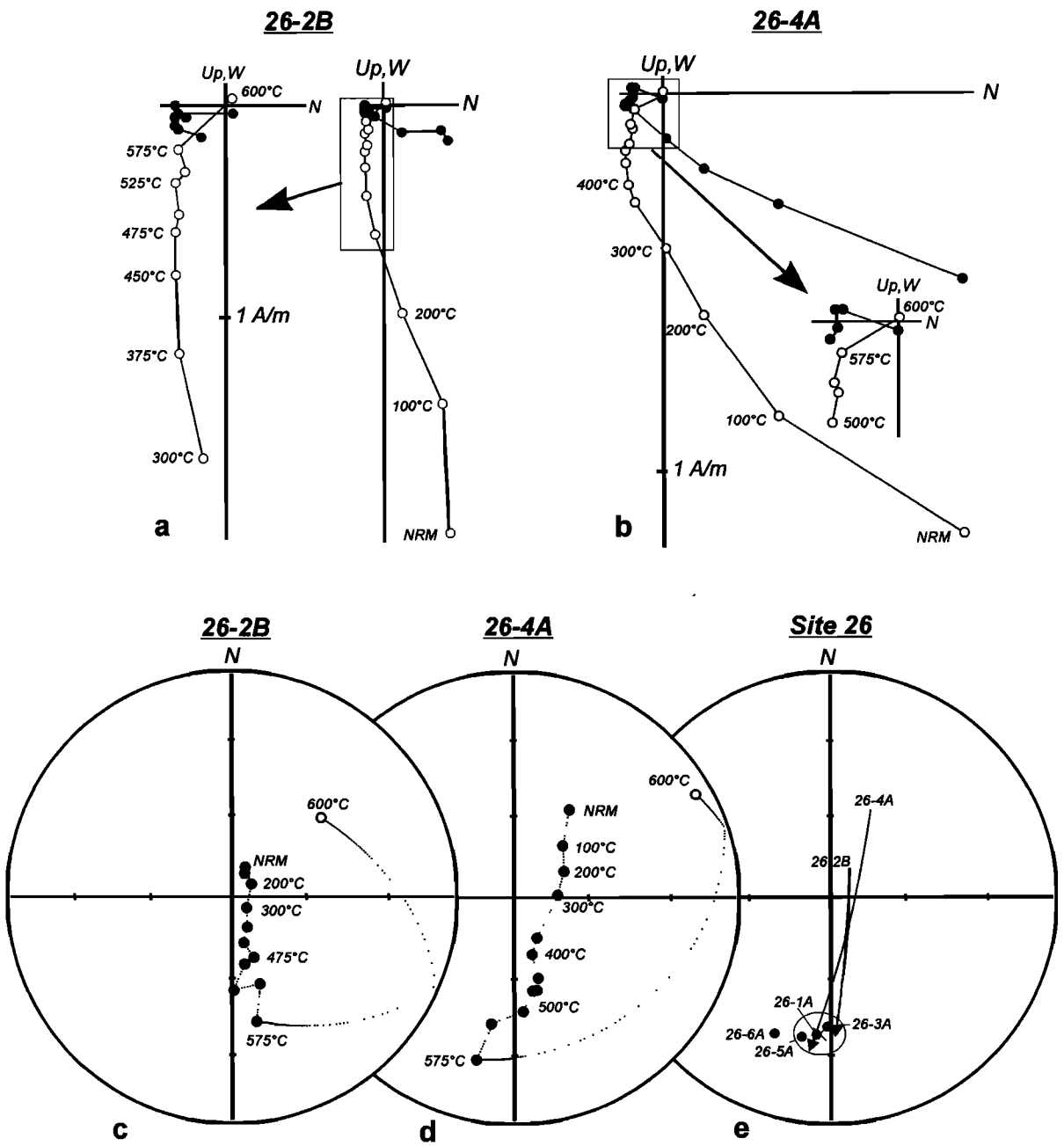


Figure 4. (a-d) Demagnetization diagrams in orthogonal [*Zijderveld*, 1967] and stereographic projection for two specimens from site 26, demonstrating almost complete overlap of the blocking temperature spectra of A and C_r. (e) Application of great circle analysis to calculate the site mean direction (orthogonal vector diagram: open and solid circles are vertical and horizontal planes, respectively; stereographic projection: open and solid circles are upper and lower hemispheres, respectively).

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ous and Permian rocks from the European Variscides [*Merabet and Guillaume*, 1988; *Perroud et al.*, 1986; *Westphal et al.*, 1987], and so B is interpreted to be a secondary magnetic overprint of late Paleozoic age.

C_r, with a southerly declination and an intermediate steep positive inclination, was isolated in 86 samples from 15 sites. It is usually carried by magnetite with maximum T_{UB} of 575°C (Figure 3c). In some specimens, C_r was stable up to 670°C, and its demagnetization curves suggest both magnetite and hematite are carriers of the magnetization (Figure 3d). C_n is present in 24 samples from five sites, is carried by magnetite (Figure 3e), hematite (Figure 3f), or both, and has a direction antiparallel to C_r.

Curved demagnetization trajectories were frequently observed in orthogonal projection (e.g., Figure 3c), indicating overlapping T_{UB} spectra of the low-temperature (LT) component A and a high temperature (HT) component (B,C_r, or C_n). In about half of these samples the overlapping spectra prevent the identification of linear demagnetization segments, requiring the use of great circle analysis. Figure 4 shows the demagnetization results for two samples from site 26 in which A can be identified in orthogonal projection up to 300°C with their directions becoming erratic above 575°C (Figures 4a and 4b). Between 300° and 575°C their remanence moves in both samples from a direction close to B on a great circle toward the direction of C_r (Figures 4c and 4d). These demagnetization circles have similar starting and ending points and are subparallel. Standard great circle intersection techniques [*Bailey and Halls,* 1984] yield mean directions at temperatures that are well below the maximum T_{UB} (Figure 4e); thus they are not considered representative for the underlying characteristic direction of C_r. Furthermore, the algorithm of *McFadden and McElhinny* [1988] failed to define the underlying characteristic direction of the natural remanent magnetization (NRM) in those sites where no stable endpoint could be identified because of the subparallel orientation of the great circles. This was the case for specimens from sites 4 and 6, which were not used for further interpretation. Similarly, the site mean directions from sites 17 to 21 were equally poorly defined, but as they were taken within few tens of meters of each other in one quarry, it is feasible to combine the results obtained here so that the number of great circles and of stable endpoint directions is sufficiently enhanced to calculate a (combined) site mean direction (Table 1).

The site mean directions are plotted for each component in Figure 5 and are listed with their resulting paleopoles in Table 2. The site means of C_r and C_n pass the reversal test of *McFadden and McElhinny* [1990] at the class C level, suggesting (1) complete removal of the LT component A, (2) sufficient time coverage of the sample set to average out any effects of secular variation, and (3) that C_r and C_n are primary.

Further evidence for the primary nature of C_r and C_n is given by a contact test on an aplitic dyke at site 16. The dyke samples have low initial remanence intensities (~10^{-4} A m^{-1}), and their demagnetization behavior became erratic above 500°–600°C. As no stable endpoints could be identified, the dyke's mean characteristic direction is defined rather poorly by great circle analysis only (Figure 6a). Nevertheless, the resulting mean direction, southerly declination and shallow upward pointing inclination, bears strong similarities to paleomagnetic direction of Permian age, described elsewhere from the Variscan Fold Belt. We also note that it is not significantly different to direction B isolated in the FIC. However, the dyke's direction is distinctly different from the C_n direction, observed in the gabbroic host rock at site 10 next to the dyke (Figure 6b), and C_r direction, observed elsewhere in the FIC. Therefore no stable magnetic overprint was acquired after the intrusion of the dyke that would conceal its primary magnetization.

5. Rock Magnetic Studies

5.1 Thermomagnetic Curves

All thermomagnetic and hysteresis measurements were done using a variable field translation balance. The thermomagnetic properties of 12 samples were initially studied to identify the magnetic carriers of A, B, C_r, and C_n. All samples show a sharp decrease of magnetization at ~580°C, the Curie temperature T_C of magnetite (Figure 7). Although hematite can be identified in many specimens during stepwise demagnetization, a decrease at the T_C of hematite at ~675°C was not observed because of insufficient saturation of the hematite phase combined with a strong paramagnetic response in both low and strong external fields, respectively.

Table 1. Paleomagnetic Data for Components B, C_r, and C_n

Site	$n/N/N_0$	In Situ		Tilt Corrected		α_{95}, deg	k
		D, deg	I, deg	D, deg	I, deg		
			Component B				
1	4/4/8	195	5	195	6	10	86
5	4/4/7	196	9	195	10	10	78
16*	4/4/6	168	-7	169	-9	28	40
23	3/3/9	179	5	179	5	15	66
29	4/4/9	178	11	177	10	13	61
			Component C_r				
1	5/4/8	197	32	194	33	14	31
2	7/6/7	223	34	221	37	10	36
3	6/5/7	223	37	220	40	9	57
4*	3/3/8	227	45	223	48	23	832
6*	3/2/8	214	39	210	42	56	226
12	6/6/8	200	30	197	31	13	30
14	7/7/7	218	34	215	37	11	36
15	5/5/5	201	30	198	32	11	61
17-21	12/12/40	189	36	186	37	8	35
22	5/5/6	195	49	190	50	12	54
23	4/4/9	213	28	210	31	15	38
24	7/7/7	206	43	201	45	9	56
25	6/6/6	216	48	211	51	6	168
26	6/5/6	188	38	185	39	9	69
28	5/5/8	211	36	208	39	6	193
29	3/3/9	182	43	179	43	16	70
30	6/6/9	203	37	199	39	10	47
			Component C_n				
8	5/5/7	15	-41	10	-42	12	44
9	5/3/6	19	-49	13	-50	11	55
10	7/7/8	5	-50	359	-51	5	130
11	6/5/6	27	-31	24	-33	4	293
13	4/4/7	3	-37	359	-37	10	147

Here n: number of paleomagnetic results from single specimens; N, number of results used for statistics; N_0, total number of cores obtained in the field; D and I, declination and inclination of the NRM; α_{95}, radius of the 95% cone of confidence about the mean direction; k, Fisher precision parameter.

* Not used for calculating mean directions of components B and C_r; see text for discussion.

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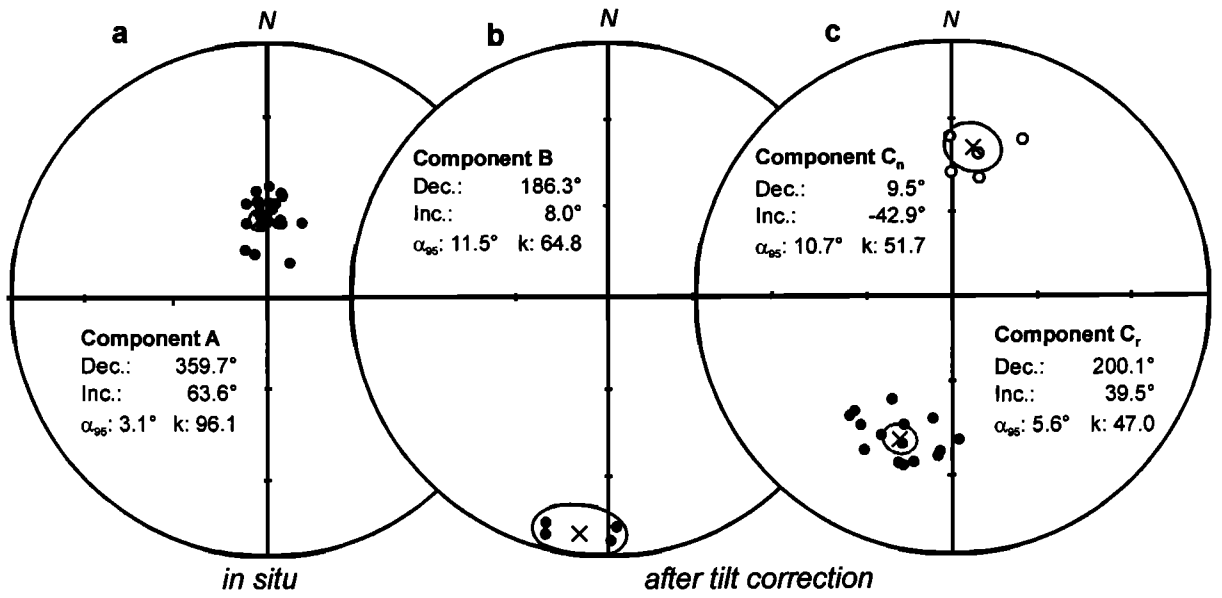


Figure 5. Site mean directions of components (a) A, (b) B, and (c) C_r and C_n in stereographic projection with their mean directions (open and solid circles are upper and lower hemispheres, respectively).

A net increase in the final magnetization was observed only occasionally after cooling, indicating that alteration and/or formation of new magnetic phases during heating is of minor importance. However, the trend of the heating and cooling curves differ in many samples. For example, the heating curve of sample 26-3 (Figure 7a) shows a wide spectrum of T_{UB} with almost linear temperature dependence and a maximum T_C of 580°C. During cooling, the range of blocking temperatures is significantly smaller, and by 400°C the magnetization has regained 90% of the final intensity. As there is no change in net magnetization before and after the experiment, chemical alteration cannot have caused the different behavior during heating and cooling. Instead, we believe the difference is the result of a multidomain (MD) effect. It is well documented that the reorganization of domain structure during cooling is an important factor in the acquisition of a thermal remanent magnetization (TRM) in MD magnetite and in the thermal history of the intensity of partial TRMs [*Shcherbakov et al.*, 1993, and references therein]. Although the thermomagnetic experiments were carried out in an external field and the magnetization is mainly induced, we propose that the difference in heating and cooling behavior is the result of comparable changes in domain structure of MD magnetite to those discussed by *Shcherbakov et al.* [1993]. For sample 12-3 this difference is significantly less, implying that here the content of MD magnetite is less (Figure 7b). The thermomagnetic behavior of most samples studied lies somewhere between those of samples 12-3 and 26-3.

5.2 Microscopic Observations

The most important magnetic carriers were identified in the samples of the FIC using reflective light microscopy. In all samples, hypidiomorphic magnetite occurs with grain sizes up to 1 mm with cleavage commonly parallel to octaeder planes. Large magnetite grains occupy up to 10 vol % in the FIC rocks, yielding a hysteresis behavior of most samples that is typical for ideal MD magnetite (see below). Small magnetite needles of <10 μm length were observed in plagioclase and amphiboles, and magnetite lamellae of various lengths occur in large ilmenite crystals, which are the result of high-temperature oxidation at >600°C [*Hargraves and Petersen*, 1971]. Because of their elongated shape the smallest magnetite phases within silicates and

Table 2. Mean Direction with Resulting Paleopole Position (South Pole) for the Components of Magnetization Identified in Rocks from the FIC

Component	N'	In Situ		Tilt Corrected		α_{95}, deg	k	PLAT	PLONG	dp, deg	dm, deg
		D, deg	I, deg	D, deg	I, deg						
A	23	359.7	63.6	-	-	3.1	69.3	86.4°S	12.0°E	3.9	4.9
B	4	-	-	186.3	8.0	11.5	64.8	36.9°S	0.9°E	5 9	11.9
C_r	15	-	-	200.1	39.5	5.6	47.0	15.5°S	11.3°W	4.0	6.7
C_n	5	-	-	9.5	-42.9	10.7	51.7	14.8°S	0.2°W	8.2	13.3
C_r+C_n*	20	-	-	198.1	40.4	4.9	45.0	15.3°S	8.6°W	3.6	5.9

N', number of site mean directions used; D and I, declination and inclination of the NRM; α_{95}, radius of the 95% cone of confidence about the mean direction; k, Fisher precision parameter; PLAT and PLONG, latitude and longitude of the paleo(south)pole; dp, dm: semiaxes of the oval of 95% confidence about the paleopole.

* The site mean directions of components C_r and C_n pass the reversal test, Class C [*McFadden and McElhinney*, 1990]

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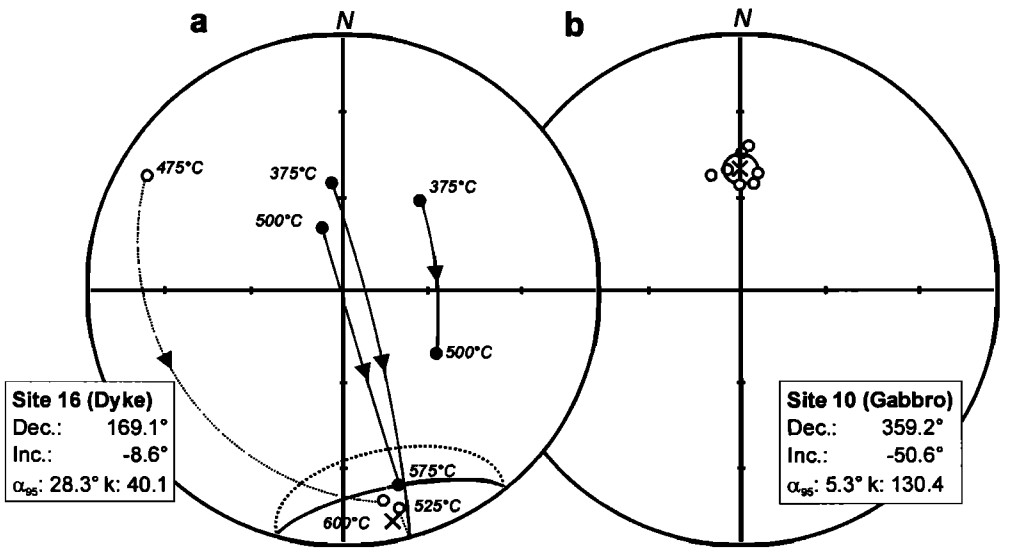


Figure 6. (a) Stereographic projection of the demagnetization great circles observed in dyke specimens at site 16 and (b) the stable endpoint direction of seven specimens of its gabbroic host rock at site 10. Symbols are as in Figure 5.

ilmenite are likely to have pseudosingle domain (PSD) structure. The presence of single domain (SD) magnetite cannot be ruled out but is beyond the resolution of standard light microscopy. A myrmecitic intergrowth of magnetite, hematite, and ilmenite was observed in large ilmenite crystals in specimen 12-3 (Figure 8). The intergrowth is thought to be the result of oxidation of hemo-ilmenite to magnetite and of the exsolution of hematite from an hemo-ilmenite host at high temperatures. These observations fit with our interpretation that the primary components of magnetizations C_r and C_n are carried by both magnetite and hematite.

In sample 1-6, hematite was identified as a rim overgrowing and partly dissolving magnetite and pyrite (Figure 9). This hematite generation is clearly of secondary origin and probably formed during the migration of fluids through the gabbro. Since B is stable up to 670°C in sample 1-6, its interpretation as a secondary magnetic overprint is strengthened by the observation of secondary hematite.

5.3 Hysteresis Properties and Grain Size Distribution of Magnetite

Remanence coercitivity H_{cr} and hysteresis parameters (H_c, J_{rs}, and J_s) were measured for 12 samples at room temperature. All samples were saturated by fields of 250 mT or lower and showed no indication of a high-coercivity magnetic phase during hysteresis. Hematite, which was identified in some samples under reflecting light and during stepwise demagnetization, is thought to be screened by the high amount of magnetite present in the samples. Under this assumption the hysteresis parameters are used to evaluate the grain size distribution of magnetite in the FIC samples. Although in a variety of samples the hysteresis properties are, at first sight, comparable to those of superparamagnetic magnetite (Figure 10a,b), we postulate MD to be predominant in these samples. This is supported by reflected light microscopy and measurements of the susceptibility at low

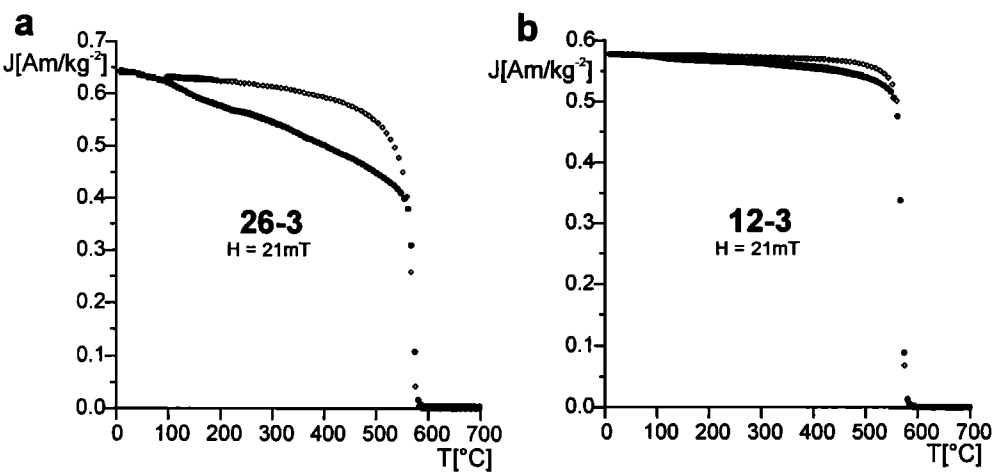


Figure 7. Thermomagnetic curves of four samples from the FIC (heating and cooling curve are given by solid circles and open diamonds, respectively) indicating magnetite as the dominant magnetic phase.

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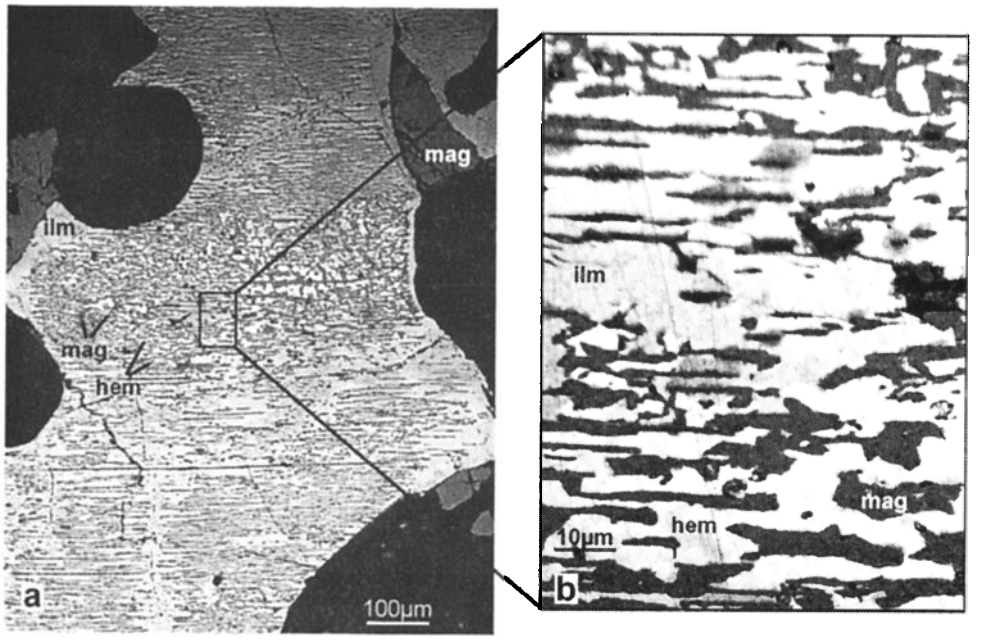


Figure 8. (a) Microphotograph of an ilmenite (ilm, light gray) crystal with magnetite (mag, dark gray) exsolution lamellae in the upper and lower parts. (b) In the central part of the ilmenite host a myrmecitic intergrowth of magnetite, hematite (hem, white), and ilmenite is present. Magnetite was identified with ferrofluid, and hematite was identified by its internal reflections under crossed polars (both figures observed with parallel polars under oil and ferrofluid; Figure 8a is 100X; Figure 8b is 1000X).

and high frequencies which do not indicate the presence of superparamagnetic particles. Only two samples show an hysteresis behavior typical for PSD magnetite (e.g., Figures 10c and 11).

Plotting H_{cr}/H_c versus J_{rs}/J_s on exponential axes [*Parry*, 1982] reveals a linear correlation of three samples (crosses and solid line in Figure 11) and its function ($H_{cr}/H_c=0.49(J_{rs}/J_s)^{-1.59}$; R=0.996) corresponds closely to the equation ($H_{cr}/H_c=0.5(J_{rs}/J_s)^{-1.6}$) for natural and artificial samples with unimodal grain size distributions [*Parry*, 1982]. Of the three samples in Figure 11 the magnetite in sample 2-5 has the smallest PSD grain sizes. The magnetic properties of a second set of samples plot above this line (solid circles and dashed line in Figure 11), and its function ($H_{cr}/H_c=0.41(J_{rs}/J_s)^{-1.30}$; R=0.968) crosses the solid line of single-sized particles at a point close to sample 2-5. This im-

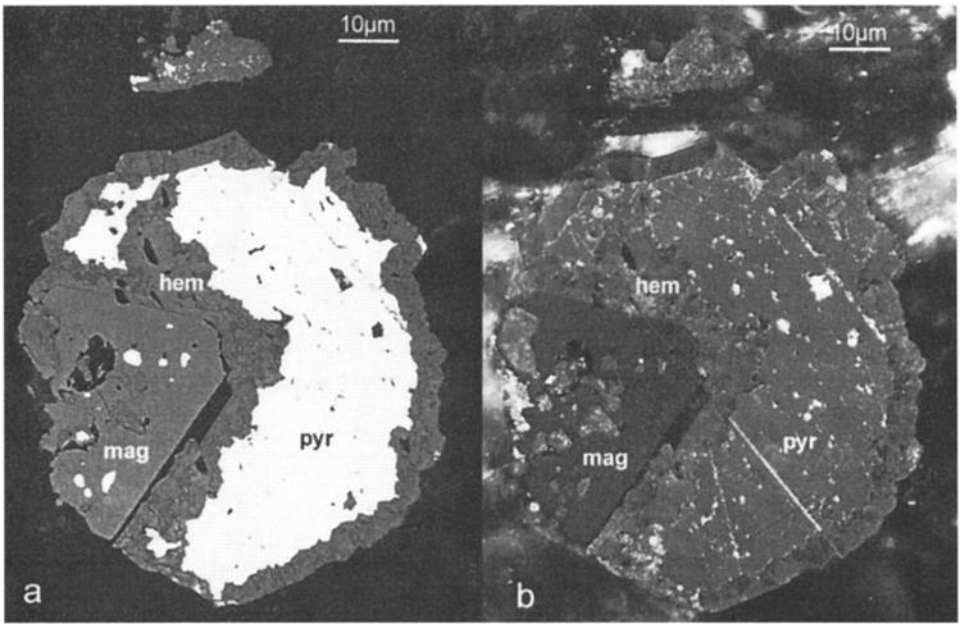


Figure 9. Microphotograph (parallel polars in Figure 9a and crossed polars in Figure 9b) of a hematite rim (hem, light gray in Figures 9a and 9b) surrounding and partly dissolving pyrite (pyr, white in Figure 9a) and magnetite (mag, dark gray in Figure 9b). Hematite was identified by its strong reddish internal reflections under crossed polars (Figures 9a and 9b are 1000X, under oil).

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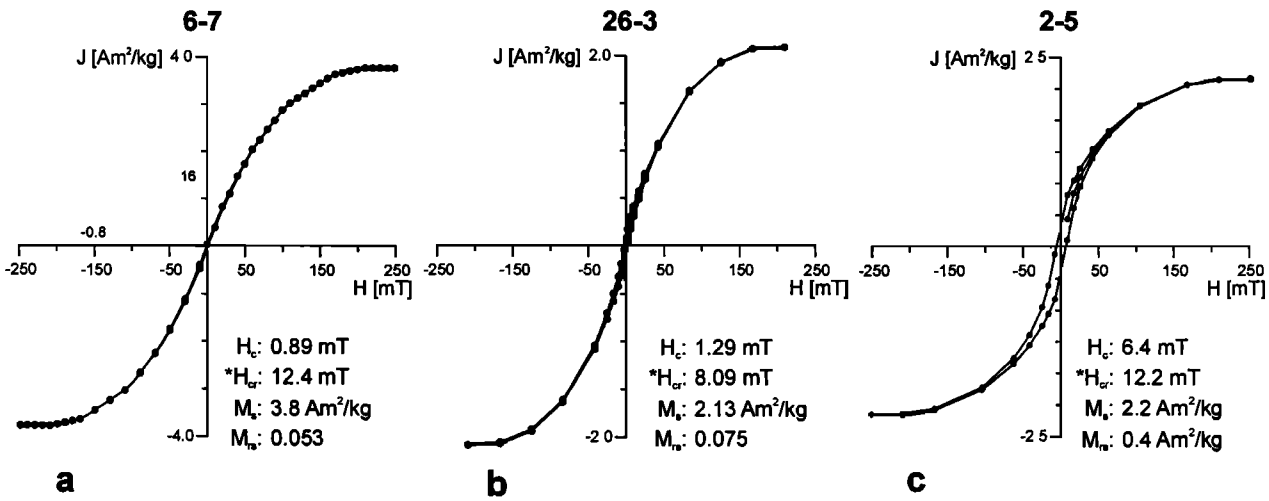


Figure 10. Hysteresis loops and parameters for three samples showing (a and b) typical behavior for multidomain (MD) magnetite and (c) pseudosingle domain (PSD) magnetite. H_{cr} was obtained using backfield measurements.

plies that sample 2-5 represents the small grain size or PSD end-member of a bimodal grain size distribution [*Parry*, 1982]. The other samples plot on the dashed correlation line in Figure 11 with increasing MD/PSD volume ratios toward higher H_{cr}/H_c ratios and with sample 6-7 having the smallest amount of PSD magnetite.

Although further rock magnetic investigations are needed to clarify the origin of the unusual high T_{UB} of the LT A component in the FIC samples, the high MD/PSD volume ratio in many samples is thought to be responsible for it and the extensive overlap of T_{UB} of A with the T_{UB} of B, C_r, and C_n. The highest MD/PSD ratio occurs in a specimen from site 6 (6-7, Figure 11) where the separation of a HT component failed in most specimens on demagnetization, owing to the extensive overlap of the T_{UB} spectra.

In a recent study on the Milton Monzonite, Sydney Basin, Australia, anomalously high T_{UB} of a LT component was explained by large intensities of a LT thermoviscous overprint component carried by MD magnetite in comparison to the HT components [*Dunlop and Özdemir*, 1997]. In FIC samples the hysteresis parameters give strong evidence for a large amount of MD magnetite, suggesting it carries A as a viscous overprint. If the intensity of B,C_r, or C_n is small compared to A because of a very small amount of PSD/SD magnetite in the sample, then the junction between the low and HT components will be pushed toward higher temperatures, and the LT component will show a quasi-exponential demagnetization curve with a tail of T_{UB} that is as high as T_C of magnetite [*Dunlop and Özdemir*, 1997].

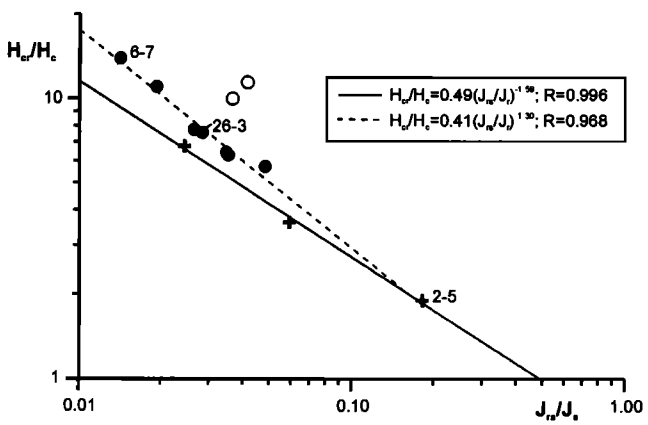


Figure 11. Exponential plot of hysteresis parameter H_{cr}/H_c versus J_{rs}/J_s after *Parry* [1982]. Two correlations fit the sample set: one (crosses and solid line) resembles the curve for samples containing unimodal grain size distributions of magnetite [*Parry*, 1982]; the second (solid circles and dashed line) magnetite is indicative for bimodal grain size distributions. Two samples (open circles) having higher ratios of H_{cr}/H_c and J_{rs}/J_s were not considered for these correlations.

6. Discussion

The paleopole obtained from the unit mean direction of component B plots at 37°S, 1°E or close to the end of the late Carboniferous segment of the apparent polar wander path (APWP) for Baltica [*Smethurst et al.*, 1998], supporting the interpretation that B is a Permo/Carboniferous overprint (Figure 12). Although remagnetization during this time is widespread in rocks from the European Variscides, its mechanism is rather poorly understood. The observation of secondary hematite rims in remagnetized samples and their scattered irregular occurrence within the FIC suggests that crystallization of hematite occured during fluid migration along discrete permeable zones, leading to the formation of a chemical remanent magnetization.

The paleopole for C_r and C_n combined at 15°S, 9°W (C_r/C_n in Figure 12) falls on the Silurian cusp of the APWP for Baltica [*Smethurst et al.*, 1998] but is displaced by 19° to the north and rotated counterclockwise by 27° away from the 365 Ma position of the APWP [*Smethurst et al.*, 1998, Table 4]. This suggests a separation between Armorica and Baltica of ~2000 km in Late Devonian times. There is geological evidence for a small oceanic basin between those two continental units in form of allochthonous mid-ocean ridge basalt (MORB)-type mafic rocks in the Rhenohercynian zone [*Franke*, 1995 and references therein], which were transported to the north during final collision of Gondwana and Baltica in the Carboniferous. However, the paleomagnetic interpretation relies heavily on the validity of the fitted APWP of *Smethurst et al.* [1998], and it should be

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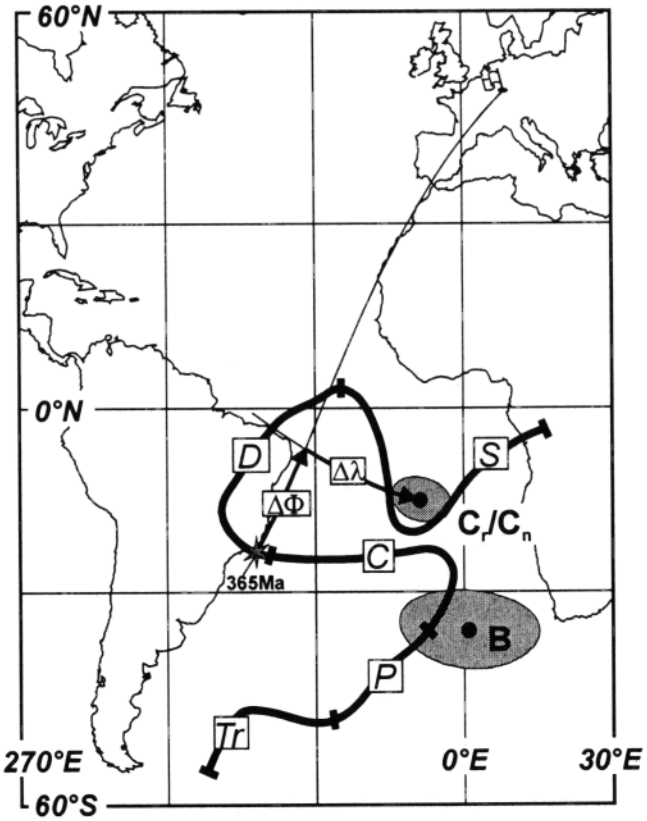


Figure 12. Paleopole positions for the unit mean directions of components B and C_r/C_n with their ovals of 95% confidence plotted against the apparent polar wander path (APWP) for stable Europe (Baltica) after *Smethurst et al.* [1998]. The arrows indicate the latitudinal ($\Delta\Phi$=19°) and longitudinal ($\Delta\lambda$=27°) differences between the 365 Ma position of the APWP (star: -23°S, 42°W [*Smethurst et al.*, 1998, Table 4]) and the paleopole position for the primary C_r/C_n magnetization of the FIC. The ages on the APWP are S, Silurian; D, Devonian; C, Carboniferous; P, Permian; Tr, Triassic.

noted that the shape of the Late Devonian segment of the path is controlled only by one paleopole from Early Carboniferous rocks from Spitzbergen [*Watts*, 1985] and one paleopole from early Devonian sediments from the Ukraine [*Smethurst and Khramov*, 1992].

The paleodeclinations of B and C_r/C_n indicate that counterclockwise rotation of the Odenwald with respect to Baltica occurred mainly before the time of acquisition of B and is ~27° in magnitude. No evidence was found for large-scale rotations with magnitudes of up to 80° as proposed by *Edel and Wickert* [1991] on the basis of paleomagnetic data from late Devonian and early Carboniferous rocks in the Odenwald, Spessart, Pfalz, and Northern Vosges. Their interpretation of paleomagnetic data from the Odenwald considers a component with a direction similar to C_r (labeled A' by *Edel and Wickert* [1991]) as secondary in origin because hematite and magnetite are its carriers. However, as shown here, reflected light microscopy clearly demonstrates that in the FIC, hematite occurs as exsolution lamellae jointly with magnetite and results from high-temperature oxidation. This observation in conjunction with both positive reversal and contact tests yield strong evidence that C_r/C_n is of primary origin.

7. Conclusions

1. The predominant magnetic mineral in the FIC is magnetite. According to the hysteresis parameters H_{cr}, H_c, J_{rs} and J_s and ore microscopy, a bimodal grain size distribution of MD and PSD magnetite occurs in most samples.

2. Hematite is also present but was not identified in hysteresis and thermomagnetic experiments. Two generations of hematite were observed under reflected light microscopy: (1) primary exsolution lamellae of hematite in ilmenite and (2) secondary alteration hematite rims on magnetite and sulfides that were probably formed during fluid migration.

3. Demagnetization experiments revealed four remanence components (A, B, C_r, and C_n), of which A occurs at low demagnetization temperatures and has a direction similar to the PDF in the sampling area. A is stable up to HT in specimens whose hysteresis properties suggest a small amount of PSD magnetite and the predominance of MD magnetite. In many such specimens, overlapping unblocking temperature spectra concealed a HT component and great care had to be taken to completely separate B, C_r, and C_n from the LT A component.

4. B is thought to be carried by the secondary hematite. It is interpreted to carry a Permo/Carboniferous overprint based on the strong resemblance of the resulting paleopole at 37°S, 1°E to the late Carboniferous/early Permian segment of the APWP for Baltica [*Smethurst et al.*, 1998].

5. C_r and C_n are both carried by primary magnetite and hematite in exsolution lamellae. They are antiparallel, and their site mean directions pass the reversal test [*McFadden and McElhinny*, 1990, Class C]. A positive contact test also indicates that C_r and C_n are primary. Their combined paleopole at 15°S, 9°W, when compared to the 365 Ma position of the APWP for Baltica [*Smethurst et al.*, 1998, Table 4], indicates a separation between Armorica and Baltica of ~2000km in Late Devonian times.

6. Counterclockwise rotation of the Odenwald with respect to Baltica of ~27° occurred mainly before the time of acquisition of component B (late Carboniferous/early Permian in age). The amount and direction of displacement of the resulting mean pole with respect to the APWP is comparable to data from the Rhenohercynian and Saxothuringian zones of the Variscides [*Bachtadse and Van der Voo*, 1986] and supports the oroclinal bending models.

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